Assignment 06
Blockchain Technology

Q1. Cryptography-
Cryptography is the practice of securing information by converting it into an unreadable format using mathematical techniques, ensuring confidentiality, integrity, and authentication.

Q2. double spending in Bitcoin-
Double spending occurs when a user tries to spend the same Bitcoin twice, exploiting digital currency's replicability. Blockchain prevents this by verifying transactions through consensus.

Q3. purpose of digital signature in blockchain-
Digital signatures ensure authenticity and integrity of transactions. They verify the sender's identity and confirm that data has not been altered.

Q4. pbft stand for in blockchain-
pbft stands for Practical Byzantine Fault Tolerance, a consensus mechanism designed to tolerate Byzantine faults in distributed

systems.

Q5. role of hashing in Blockchain-
Hashing secures data by converting it into a fixed-length string. It links blocks together, prevents tampering, and ensures immutability of the ledger.

Q6 function of a Bitcoin wallet-
A Bitcoin wallet stores private keys and allows users to send, receive, and manage Bitcoin securely.

Q7 PoW and PoS -
Proof of Work (PoW) Miners solve complex puzzles, energy-intensive, used in Bitcoin.

Proof of Stake (PoS) Validators chosen based on stake, energy-efficient, used in Ethereum 2.0.

Q8 Types of Cryptography in Blockchain -
Symmetric Cryptography Same key for encryption and decryption.

Asymmetric Cryptography Public-private key

pairs

Hash Functions: One-way functions for integrity

Q7. hashing works in Blockchain-
Hashing converts input data into a fixed-length output using algorithms like SHA-256
Example

text
Input: "Hello Blockchain"
SHA-256 Output: 5d41402abc4b2a76b9719d911017c592
Ensures immutability

Links blocks via previous block hash

Detects tampering instantly

Q10. Difference between Pseudo-anonymity and Anonymity-
Pseudo-anonymity: Users are identified by addresses (e.g. Bitcoin wallet address). Transactions are traceable but not directly linked to

identity

Anonymity No traceable identity, complete privacy

All Technologies used in Blockchain
Cryptography Secures transactions

Distributed Ledger Technology (DLT) Shared immutable records

Consensus Mechanisms PoW, PoS, PBFT

Smart Contracts Self-executing agreements

Peer-to-Peer Networks Decentralized communication

Hashing Algorithms Ensure immutability

All Forking in Blockchain & Challenges of Bitcoin
Forking Types
Soft Fork Backward-compatible changes
Hard Fork Permanent divergence, requires all nodes to upgrade
Challenges in Bitcoin
Scalability